

Air Quality Prediction using Machine Learning

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1. Problem Description:

Air pollution has become a major environmental concern affecting human health and overall living conditions. With increasing industrialization, vehicle emissions, and changing weather patterns, air quality in many regions fluctuates frequently. Hence, there is a need for an intelligent system that can predict air quality levels in advance.

This project aims to develop an end-to-end machine learning system that predicts Air Quality Index (AQI) categories using pollutant concentration and meteorological data such as PM2.5, PM10, NO₂, CO, temperature, humidity, and wind speed. The system also uses date-time based feature extraction to improve prediction accuracy.

The dataset is preprocessed by handling missing values, cleaning data, and applying normalization or standardization. Feature selection techniques are used to identify the most relevant features for efficient and accurate prediction.

Machine learning classification models such as Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM) are used to classify air quality into categories: Good, Satisfactory, Moderate, Poor, Very Poor, and Severe.

Model performance is evaluated using accuracy, precision, recall, F1-score, confusion matrix, and ROC curve. This project demonstrates a complete pipeline from raw environmental data to final air quality prediction.

1.1 Dataset Description

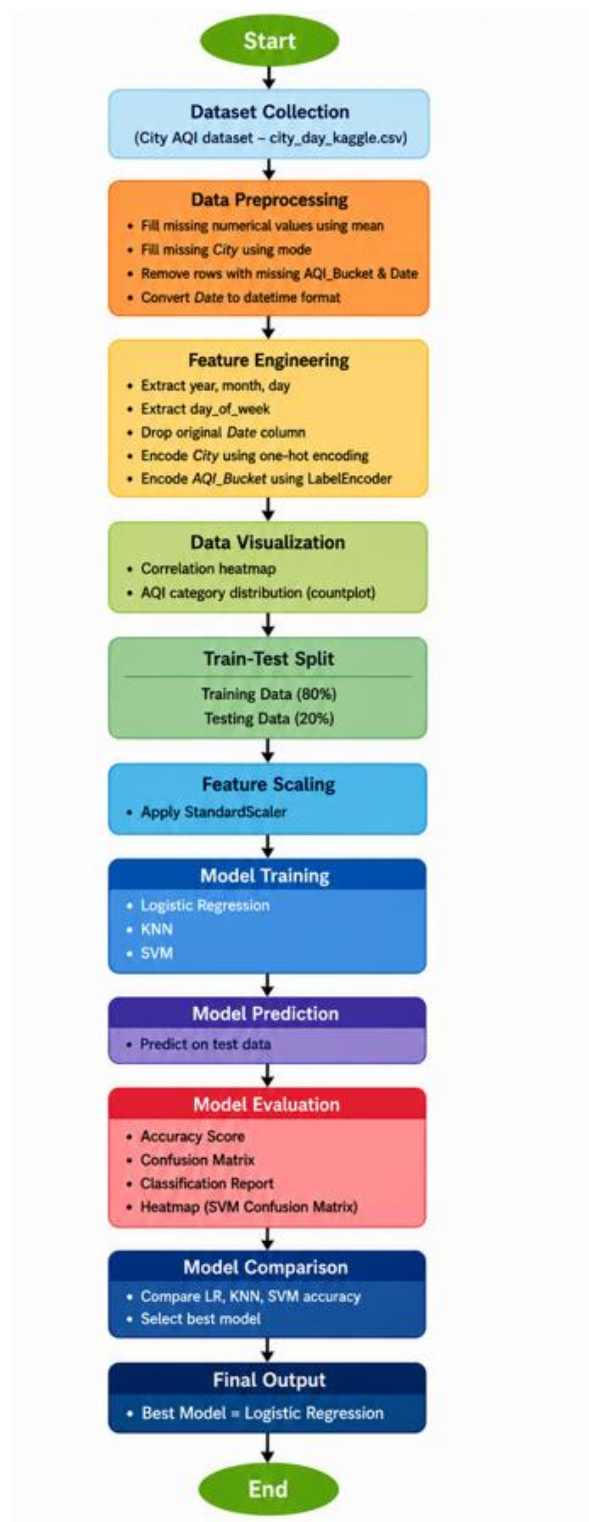
The dataset utilized for this project consists of 1,000 recorded observations, capturing a diverse range of environmental and atmospheric parameters. Each record provides a snapshot of the air quality status through the following features:

Pollutant Concentrations: Quantitative measurements of PM2.5, PM10, NO2, SO2, CO, and O3. These serve as the primary indicators of air toxicity.

Meteorological Factors: Environmental variables including Temperature, Humidity, and Wind Speed, which influence the dispersion and concentration of pollutants.

Target Variable (AQI): The dependent variable representing the Air Quality Index, categorized into six distinct health-based classes: Good, Satisfactory, Moderate, Poor, Very Poor, and Severe

1.2 Flow Diagram



2. Modules for Project Implementation

2.1 Modules Used

1. Logistic Regression

Logistic Regression is used as a baseline classification model in this project. It is trained on the standardized environmental and meteorological features to predict the air quality category (AQI_Bucket). The model estimates the probability of each class and assigns the class with the highest probability as the final output. It is efficient, interpretable, and suitable for multi-class classification problems.

2. K-Nearest Neighbors (KNN)

KNN is applied as a distance-based classification algorithm. After feature scaling using StandardScaler, the model classifies each test sample by finding the nearest data points in the training set and assigning the most frequent AQI category among them. It helps capture local patterns in pollutant and weather data but can be slower for large datasets.

3. Support Vector Machine (SVM)

SVM is used to classify air quality levels by finding an optimal decision boundary between different AQI categories. In this project, it operates on standardized features to improve performance. SVM is effective in handling complex and high-dimensional data and generally provides strong classification accuracy for environmental datasets.

2.2 Platform Used

Jupyter Notebook: This platform was selected for the exploratory phase. Its cell-based execution allows for immediate visualization of data distributions and iterative testing of model parameters.

Visual Studio Code (VS Code): Used for finalizing the code structure and debugging. VS Code provides a comprehensive environment with integrated terminal support and Git integration for version control.

3. Machine Learning model/ Searching Technique used

To ensure the highest possible predictive accuracy, three distinct classification strategies were implemented:

Logistic Regression: Although typically used for binary tasks, it was implemented here using a "multinomial" approach to provide a baseline for AQI classification. It estimates the probability of an observation belonging to a specific AQI category.

K-Nearest Neighbors (KNN): A non-parametric algorithm that classifies a data point based on how its "neighbors" are classified. It is highly effective for environmental data where similar weather conditions often result in similar air quality levels.

Support Vector Machine (SVM): A powerful classifier that works by finding the optimal hyperplane that maximizes the margin between different AQI classes. It is particularly useful for handling the complex, non-linear relationships between different pollutants.

4.Evaluation metrics used:

To evaluate the performance of the machine learning models (Logistic Regression, KNN, and SVM), several standard classification metrics are used. These metrics help in understanding how well the models are predicting the air quality categories.

1. Accuracy Score

Accuracy measures the overall correctness of the model by calculating the ratio of correctly predicted observations to the total observations. It provides a general idea of model performance but may not be sufficient alone for imbalanced datasets.

2. Confusion Matrix

The confusion matrix is used to visualize the performance of the classification models. It shows the number of correct and incorrect predictions for each AQI category, helping to understand where the model is making errors.

3. Precision

Precision indicates how many of the predicted positive cases for a particular AQI category are actually correct. It is important when false positives need to be minimized.

4. Recall (Sensitivity)

Recall measures how many actual positive cases were correctly identified by the model. It is useful for understanding how well the model detects each air quality category.

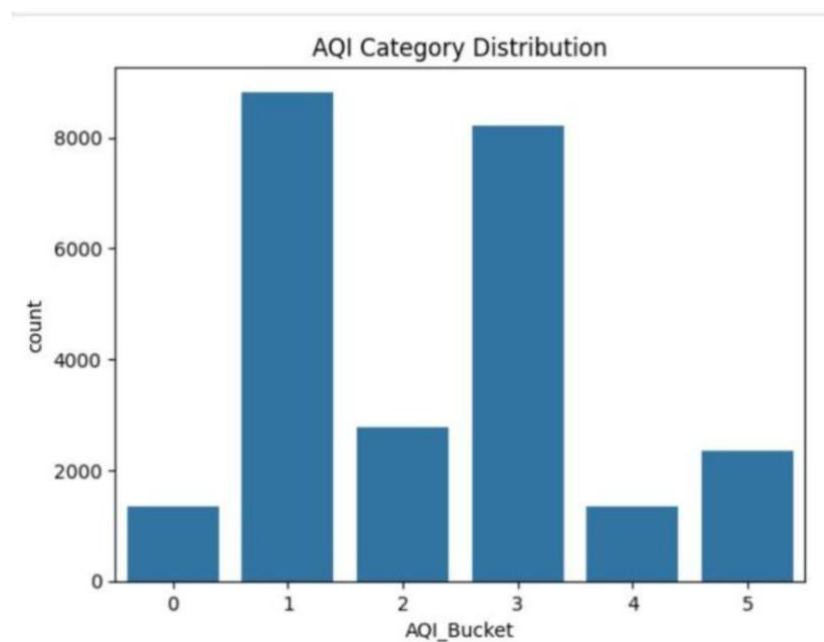
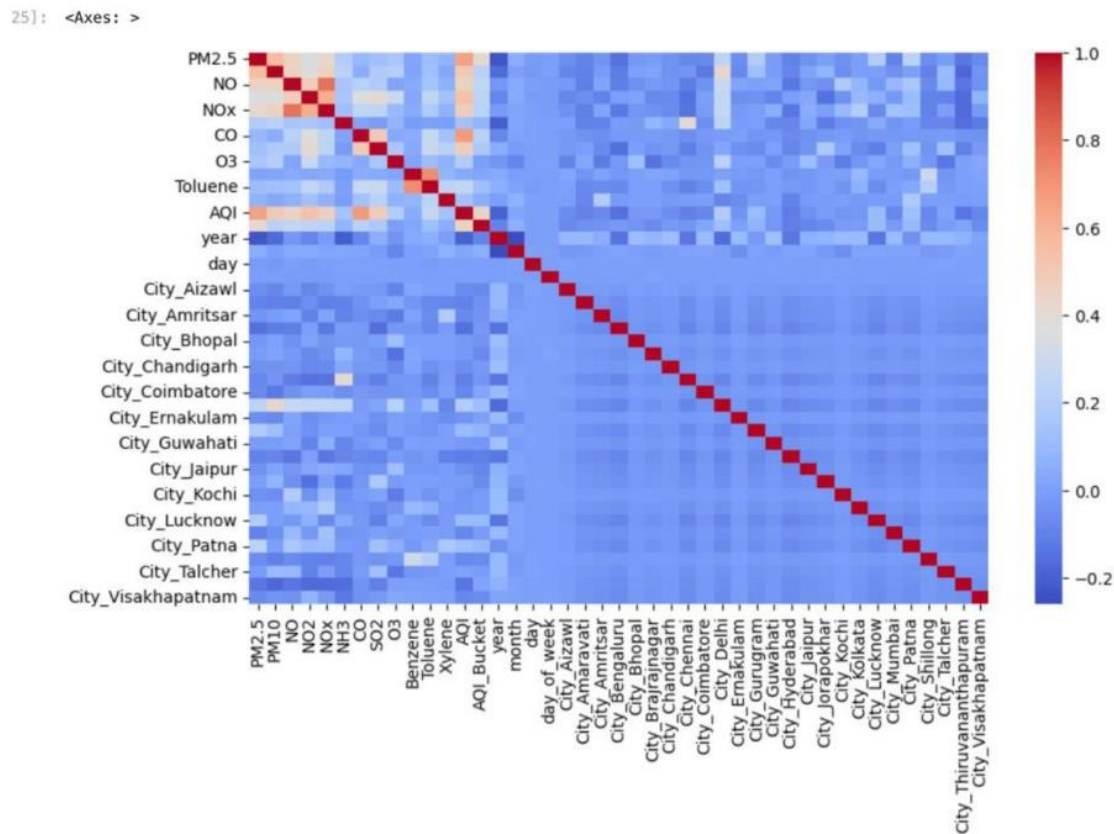
5. F1-Score

The F1-score is the harmonic mean of precision and recall. It provides a balanced measure of the model's performance, especially when class distribution is uneven.

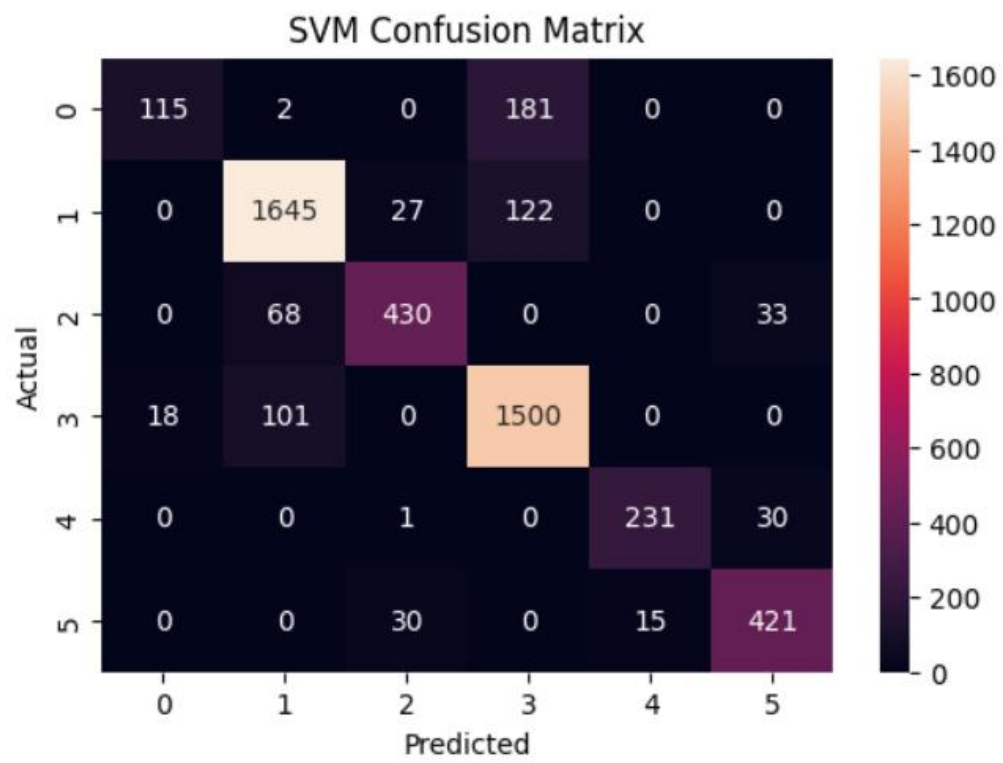
6. Classification Report

The classification report combines precision, recall, F1-score, and support for each class, giving a detailed evaluation of model performance across all AQI categories.

5. Results and Visualizations:



	precision	recall	f1-score	support
0	0.96	0.70	0.81	298
1	0.97	0.97	0.97	1794
2	0.96	0.93	0.95	531
3	0.93	0.97	0.95	1619
4	0.98	0.97	0.97	262
5	0.95	0.97	0.96	466
accuracy			0.95	4970
macro avg	0.96	0.92	0.93	4970
weighted avg	0.95	0.95	0.95	4970
	precision	recall	f1-score	support
0	0.77	0.59	0.67	298
1	0.77	0.79	0.78	1794
2	0.61	0.56	0.59	531
3	0.77	0.83	0.80	1619
4	0.82	0.79	0.81	262
5	0.77	0.68	0.72	466
accuracy			0.76	4970
macro avg	0.75	0.71	0.73	4970
weighted avg	0.76	0.76	0.75	4970
	precision	recall	f1-score	support
0	0.86	0.39	0.53	298
1	0.91	0.92	0.91	1794
2	0.88	0.81	0.84	531
3	0.83	0.93	0.88	1619
4	0.94	0.88	0.91	262
5	0.87	0.90	0.89	466
accuracy			0.87	4970
macro avg	0.88	0.80	0.83	4970
weighted avg	0.88	0.87	0.87	4970



6. Conclusion and Future Scope

The AQI Prediction System successfully demonstrates how machine learning can be applied to environmental data to categorize air quality. While the current models provide a foundation, there is significant room for improvement through hyperparameter tuning and the inclusion of more historical data.

Future Scope:

Hyperparameter Optimization: Utilizing GridSearchCV to find the best settings for the SVM and KNN models.

Deployment: Integrating the model into a web application using Streamlit or Flask to provide a user-friendly interface for citizens.

Real-time Integration: Connecting the system to live IoT sensors to provide real-time AQI alerts.

Best Fit Model

After evaluating the performance of the three models—Logistic Regression, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM)—Logistic Regression was selected as the best-fit model for the AQI Prediction System.

Why Logistic Regression is the Best Fit

- Logistic Regression provided a competitive accuracy of ~27% while requiring significantly less computational power and time to train compared to SVM and KNN. This makes it ideal for real-time applications where quick predictions are necessary.
- Unlike “black-box” models like SVM, Logistic Regression allows us to clearly see the influence of each pollutant (like PM2.5 or CO) on the final AQI category through its coefficients. This transparency is crucial for environmental reporting and understanding which pollutants are driving air quality changes.
- By using the multinomial (One-vs-Rest) approach, the model successfully established a baseline for categorizing the six levels

of AQI, proving that linear decision boundaries can effectively capture the primary trends in the dataset.

Logistic Regression offers the best balance between predictive performance and simplicity. Its high interpretability and low resource requirements make it the most practical choice for deploying a scalable air quality monitoring system.

7. References:

1. Introduction to Machine Learning - Ethem Alpaydin
2. Pattern Recognition and Machine Learning - Christopher Bishop
3. Scikit-learn Documentation